

CS 473, Spring 2023

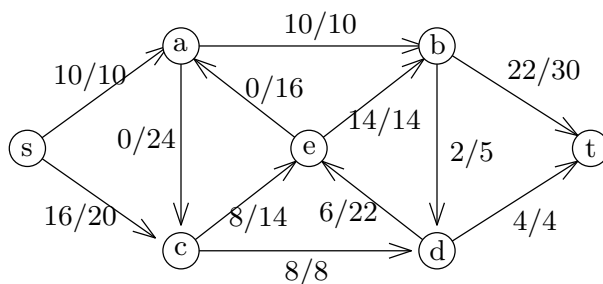
Midterm Exam 2

Name:	
NetID:	

- Date/Time: Apr 3, 2023 (Monday), 7:00pm–9:00pm
- You may bring **one double-sided 8.5" × 11" cheat sheet**, with your name and NetID written on the upper right corner. (Two single-sided sheets are okay.) The sheet should be **handwritten** by you (not photocopied or printed).
- Except for your writing implements, your cheat sheet, and your university ID, please put everything away for the duration of the exam. In particular, please turn off and put away *all electronic devices* other than those that are medically necessary.
- This exam booklet is **double-sided!** There are 4 questions (some of which may have multiple parts). The total number of points is 100.
- **At the top of *every* page, please write your name.** (We will scan every page in your exam booklet into Gradescope.)
- All answers should be placed in the spaces given. If you need more space to complete an answer, you may use the blank pages at the end, but please clearly indicate where we should look.
- You may use the blank pages at the end for scratch work.
- Submit your cheat sheet (and all scratch papers) along with your exam booklet when you are finished.
 - If you are NOT using a cheat sheet, please check this box. ☐
- You are reminded about the course's, the department's, and the university's academic integrity policies (the penalty for cheating is severe).
- All exams are strictly confidential for at least 24 hours, or until all conflict exams have been taken. Do not discuss your exam with anyone, either in person or online.

1. [25 PTS] *Short questions.*

- (a) [10 PTS] Run *one iteration* of the Ford–Fulkerson algorithm for the maximum flow problem on the following example, starting with the initial flow shown. Here, the x/y label on an edge indicates that the edge has capacity y and initial flow value x . Show the residual graph G_f , and the augmenting path chosen (if there are choices, pick any you prefer), and the new flow. Is the new flow after one iteration optimal? If so, provide a minimum s - t cut with capacity equal to the max flow value.



Show the residual graph G_f and your augmenting path:

Show the new flow:

Is the new flow optimal? If so, provide a corresponding minimum s - t cut:

- (b) [4 PTS] Consider an algorithm A whose input is an integer. Suppose that for any N , algorithm A works correctly for $N/4$ of the input numbers $\{1, \dots, N\}$ and is incorrect for $3N/4$ of the input numbers $\{1, \dots, N\}$. Does it necessarily follow that A is a randomized Monte Carlo algorithm with error probability $3/4$? Explain.
- (c) [6 PTS] Suppose there is an efficient Monte Carlo algorithm A that solves a problem in $T_A(n)$ time with error probability at most 0.1 . Suppose there is an efficient (deterministic) algorithm V that can efficiently verify whether an answer is correct in $T_V(n)$ time. Briefly describe how one can obtain a Las Vegas algorithm to solve the problem, and analyze its expected running time.
- (d) [5 PTS] True or False: In the Hungarian method for maximum matching in an unweighted bipartite graph with n vertices, the number of iterations needed is always at most n . Briefly explain.

2. [26 PTS] We are given two strings $u = a_1a_2 \cdots a_n$ and $v = b_1b_2 \cdots b_n$ of length n . We are also given a number $\ell \leq n$. We want to find a string of length ℓ that is simultaneously a substring of u and v , or determine that no such string exists. (Recall that substrings need to be contiguous, unlike subsequences.)
- (a) [23 PTS] Design and analyze a Las Vegas algorithm to solve this problem in $O(n)$ expected time, by using the Karp–Rabin fingerprinting technique.
(Recall the fingerprint function from class: $F_x(a_1a_2 \cdots a_k) = (a_1x^{k-1} + a_2x^{k-2} + \cdots + a_k) \bmod p$ for a fixed prime p . You may also use hashing as a subroutine.)

Pseudo-code:

Analysis of expected running time: (specify your choice of p)

- (b) [3 PTS] Using part (a),¹ briefly describe how to find the *longest* common substring of u and v . What is the final expected running time?

¹As usual, you can still do part (b) even if you are unable to do part (a) correctly.

3. [23 PTS] Consider the following problem (one of the numerous variations of “coin weighing puzzles”): We are given a set of n coins. Most of the coins are *genuine* and have the same weight, say, 1 unit, but there may be up to 2 *fake* coins, which have weight different from 1 unit. We have a scale that can weigh an any given subset of coins, i.e., determine the exact sum of the weights in the subset. The problem is to decide whether there are any fake coins (you are *not* required to find them, but just decide existence).

[What makes the problem nontrivial is that the adversary/counterfeiter could create 1 fake coin of weight $1 + \delta$ and another of weight $1 - \delta$, so that we would not be able to detect the existence of fake coins if we simply weigh the whole set (the scale would report total weight n).]

Design and analyze a Monte Carlo algorithm that solves the problem using $O(1)$ weighings with error probability at most 0.01.

[Bonus (3 pts), if your analysis works even when there could be an arbitrary number of fake coins.]

Pseudo-code or algorithm description:

Analysis of the number of weighings:

Write Your Name: _____

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Analysis of error probability:

4. [26 PTS] We are given a bipartite graph $G = (V, E)$, where V is divided into V_L (the left part) and V_R (the right part). We are also given subsets $S_L \subset V_L$ and $S_R \subset V_R$. Given a number k , we want to decide whether there exists a matching in G that *uses exactly k edges and is required to cover all vertices in S_L and in S_R* (it may cover additional vertices).

Describe a polynomial-time algorithm to solve this problem by constructing a directed graph (flow network) G' and running a maximum-flow algorithm on G' .

(Hint: modify the standard reduction from bipartite matching to max flow, by adding some extra vertices/edges to G' .)

Description of your reduction and construction of G' (and edge capacities):

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Running time analysis:

Correctness proof:

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